



LESSON 5.10

SOLVING REAL-WORLD PROBLEMS ON A COORDINATE PLANE WITH THE PYTHAGOREAN THEOREM

LESSON INTRODUCTION

MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

LEARNING TARGETS

- I can apply the Pythagorean theorem to determine the distance between two points on the coordinate plane.
- I can apply the Pythagorean theorem in a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

WORKING TOWARD

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

ESSENTIAL QUESTIONS

- How can I find the distance between two points on the coordinate plane using what I know about the Pythagorean theorem?
- How can I apply the Pythagorean theorem to solve problems in real-world context?

LESSON NARRATIVE

In this lesson, students practice applying what they have learned about the Pythagorean theorem to real-world contexts. Students start by activating prior knowledge of solving for the hypotenuse of a triangle on the coordinate plane. Students then engage in a Collaboration where they use critical thinking to solve real-world problems involving the Pythagorean theorem. Ultimately, students play a game to practice what they have learned.

COMPONENT SUMMARY

5.10.1 WARM-UP (~5 MIN.)

Activate prior knowledge on Pythagorean theorem

5.10.3 ACTIVITY (20 MIN.)

Activity applying distance and Pythagorean theorem

5.10.5 WRAP-UP (~5 MIN.)

Error analysis of student work

5.10.2 COLLABORATION (15 MIN.)

Critical-thinking activity

5.10.4 YOUR TURN! (5 MIN.)

Practice applying the Pythagorean theorem on the coordinate plane

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Pythagorean theorem
- Distance
- Coordinate plane

MATERIALS

- Calculators
- (Optional) Highlighters
- (Optional) Ruler or straightedge

ADDITIONAL PREPARATION

- This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (word wall, math cards, etc.).

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may substitute the numbers in the incorrect spot in the Pythagorean theorem.
- Students may make errors when finding the vertical or horizontal distances on the coordinate plane.

THINGS TO CONSIDER

- Consider providing students with straightedges to help draw line segments.
- If time is an issue, then consider assigning 5.10.4 Your Turn! for homework.
- During the 5.10.2 Collaboration and/or 5.10.3 Activity, consider working with a small group who may need further support on the concepts learned in the unit.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 5.10.2 Collaboration, consider implementing a Think-Pair-Share for question 1e. Students can compare the actual times they calculated with their predictions before discussing with their partner. Have students share out how close they were to the actual time. For any students who were very close, how did they make their prediction

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

This is the final lesson in this unit and engages students in several opportunities for collaboration. Partner discussion allows for mathematical discourse to help students synthesize the content learned throughout this unit. Encourage students to verbalize their strategies as they participate in 5.10.3 Activity.

DIFFERENTIATE INSTRUCTION

For 5.10.2 Collaboration, for a kinesthetic approach, consider recreating this example in real time. Have students stand in the place of each player on a 3D coordinate plane. For students who are struggling to make a prediction using the diagram, a more kinesthetic approach may help.

5.10.1 WARM-UP: BELL WORK

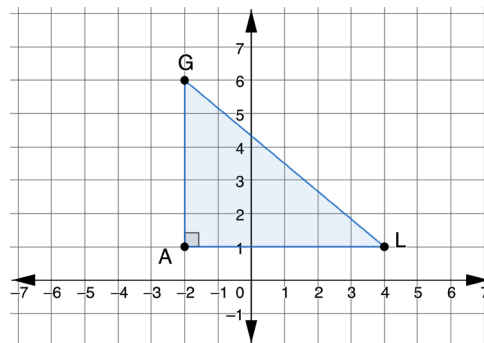
TEACHER MOVES

The information provided from the students here, coupled with what was observed during instruction, can provide information for which students may need further support through small-group instruction.



5.10.1 Warm-Up: Bell Work

Triangle GLA is shown on the coordinate plane.



- Find the exact length of $\overline{GL}$.
 $\sqrt{61}$ units



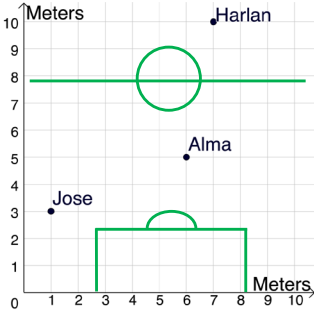
5.10.2 Collaboration: Pass the Ball!

1. Work with your partner to answer the following questions.

The figure shows the position of different soccer players during a recent game. Jose wants to pass the ball to Harlan.

He has two options.

- A. Pass the ball directly to Harlan.
- B. Pass the ball to Alma, who can pass the ball to Harlan.



- a. Which strategy, A or B, do you think will get the ball to Harlan faster?
Answers will vary. I think strategy A will get the ball to Harlan faster. I think strategy B will get the ball to Harlan faster.
- b. How far does the ball travel with strategy A?
 $\sqrt{85}$ meters (9.2)

- c. How far does the ball travel with strategy B?
 $\sqrt{29} + \sqrt{26}$ meters (10.5)

Jose and Alma pass the ball at different speeds. Jose can kick the ball forward at a rate of 2 meters per second, while Alma can kick the ball forward at a rate of 4 meters per second.

- d. Calculate the time that each strategy would take.

$$\frac{\text{Distance}}{\text{Rate}} = \text{Time}$$

Jose to Harlan	Jose to Alma	Alma to Harlan
<i>4.6 seconds</i>	<i>2.7 seconds</i>	<i>1.3 seconds</i>
Total Time for Strategy A: <i>4.6 seconds</i>	Total Time for Strategy B: <i>4 seconds</i>	

- e. How did the actual times for each strategy compare to your prediction?
Answers will vary. The actual time matched my prediction for the faster strategy. The actual time did not match my prediction for the faster strategy.

5.10.2 COLLABORATION: PASS THE BALL!

DIFFERENTIATE INSTRUCTION

Consider using information gathered from previous lessons on students' mastery of the Pythagorean theorem to create strategic groupings of students.

ENRICHMENT

This section requires students to work through several steps to be able to find the distance the ball would travel. Students must create three right triangles using the points given and apply the Pythagorean theorem to each to find the diagonal distances to answer questions 1b and 1c. Break down this question for students who are struggling to get started.

- What information is given? How can that be used to find missing side lengths?
- How can you create right triangles to help solve this problem? How many triangles need to be created to accurately solve for the distance between points?

DIFFERENTIATE INSTRUCTION

Consider checking students' work/answers for questions 1b and 1c before students move on to question 1d. If students have the incorrect answer for distance, it will affect their ability to find the time.

TEACHER MOVES

Think-Pair-Share

Consider implementing a Think-Pair-Share for question 1e. Students can compare the actual times they calculated with their predictions before discussing with their partner. Have students share out how close they were to the actual time. For any students who were very close, how did they make their prediction?

5.10.3 ACTIVITY: TARGET PRACTICE

TEACHER MOVES

Circulate as students work with their partners. Continue to remind students of the steps needed to find the distance. Consider posting the steps somewhere all students can reference as they engage in this activity.

ENRICHMENT

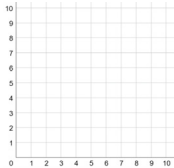
As you circulate, consider posing questions to encourage student thinking and prompt them to utilize strategies learned throughout the unit. How could you use the skills we have learned to help you find the location of your partner’s point? What other information would have been helpful for you to know?



5.10.3 Activity: Target Practice

You and your partner will attempt to guess the coordinates of a point on each other’s graphs. If the guess is incorrect, the owner of the board will calculate the distance between the two points, then share the distance with the guesser. Take turns guessing the coordinates of each other’s points and calculating the distance between the two. The first partner to guess the coordinate correctly or with the closest distance after 4 rounds, wins!

Answers will vary.



- Plot an ordered pair on the grid (do not let your partner see it).
- Record your partner’s guesses and your guesses in the tables below.

Round	Your Point	Partner’s Guess	Distance Calculations
1			
2			
3			
Final Guess			

Round	Your Guess	Distance From Partner’s Point
1		
2		
3		
Final Guess		

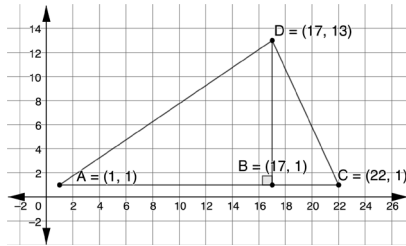
- What strategies did you use to help determine your next guess?

Answers will vary.



5.10.4 Your Turn!

- Triangle ABD is shown on the coordinate plane.



Find the perimeter of Triangle ABD .

48 units

- A walkway is constructed around a triangular field, as shown.



How far is the walk around the field? Round your answer to the nearest tenth.

206 meters

5.10.4 YOUR TURN!

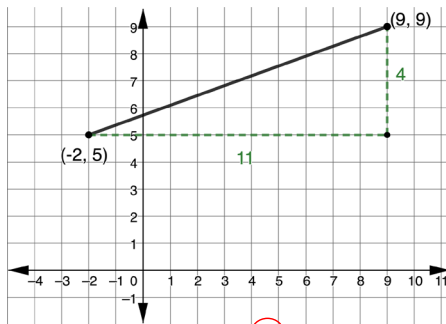
TEACHER MOVES

If you notice students struggling to plug the lengths into the formula to find the hypotenuse, prompt students to highlight and label the hypotenuse and legs on the triangle as they correspond to a , b , or c in the Pythagorean theorem.



5.10.5 Wrap-Up: Error Analysis

Ellena found the distance between the points $(-2, 5)$ and $(9, 9)$. Her work is shown.



$$\begin{aligned} 4^2 + b^2 &= 11^2 \\ 16 + b^2 &= 121 \\ b^2 &= 105 \\ b &= \sqrt{105} \end{aligned}$$

The distance is $\sqrt{105}$ units.

- Circle the error(s) made.
- Write a short explanation to Ellena explaining the mistake and what she needs to remember about determining the distance between two points on a coordinate plane.

Answers will vary. Your mistake is that you did not substitute 11 into the b^2 of the Pythagorean Theorem. Remember, b is the leg of the triangle and c is the hypotenuse.

5.10.5 WRAP-UP: ERROR ANALYSIS

DIFFERENTIATE INSTRUCTION

To differentiate this Wrap-Up, have students who may struggle to express their thinking in writing correct their errors through mathematical representations rather than words. Alternatively, consider allowing students to record their responses orally to provide an additional entry point.